

1.3 - StorageCompression - Notes.pdf

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Unit 1.3 – Data Storage & Compression

Candidates should be able to:

- Understand how data storage is measured. (From bit to exbibyte).
- Calculate the file size of an image file and a sound file, using information given
- Understand the purpose of and need for data compression
- Understand how files are compressed using lossy and lossless compression methods

Units of Data

The smallest unit of data we can currently store (we're not at quantum computing yet!) is the bit (binary digit), which can represent either 0 or 1.

Different numbers of bits can be grouped together to form larger units of data. Many of these will already be familiar to you from your everyday usage of electronic devices.

Unit	Size
Bit	1 bit
Nibble	4 bits
Byte	8 bits
Word	2 bytes
Kilobyte (KB)	1000 bytes
Megabyte (MB)	1000 KB
Gigabyte (GB)	1000 MB
Terabyte (TB)	1000 GB
Petabyte (PB)	1000 TB
Exabyte (EB)	1000 PB
Zettabyte (ZB)	1000 EB
Yottabyte (YB)	1000 ZB

Unit	Size
Bit	1 bit
Nibble	4 bits
Byte	8 bits
Word	2 bytes
Kibibyte (KiB)	1024 bytes
Mebibyte (MiB)	1024 KiB
Gibibyte (GiB)	1024 MiB
Tebibyte (TiB)	1024 GiB
Pebibyte (PiB)	1024 TiB
Exbibyte (EiB)	1024 PiB
Zebibyte (ZiB)	1024 EiB
Yobibyte (YiB)	1024 ZiB

Megabytes or Mebibytes?

- 1MB = 1000 kilobytes (1000³ bytes)
- 1MiB = 1024 kibibytes (1024³ bytes)

For this course, you should understand and be familiar with both!

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Calculating File Sizes

Calculation of file sizes has already been covered in the previous units:

- 1.2.2 – Sound
- 1.2.3 – Images

Please refer to the notes, presentations and other resources for those topics.

File Compression

All types of files take up space in memory. Compression is the process of reducing the file size in memory. The effects of file compression are:

- Less bandwidth is required to transmit files
- Less storage space is required to store files
- Less time is required to transmit files



There are two (fairly self-explanatory) types of compression: **Lossy** and **Lossless**.

Lossless compression:

Lossless compression allows the file size to be reduced **without losing any of the original data**, meaning that the original file can be re-created exactly from the compressed file. It works by searching for patterns in the file so, instead of repeatedly storing a block of identical data, the data is stored once and then indexed. Further occurrences are simply stored as the index number so the decompression software can simply look up the data and place it back in the correct position. Example lossless formats: .zip, .png, .tif, .svg

Lossy compression:

Files that include a lot of unique information, such as bitmap graphics, audio or video files, cannot be compressed much with lossless compression because there is so little repeated data. Lossy compression **removes data that is not needed**, either because a drop in quality is acceptable or because the difference cannot be detected by the human eye or ear.

With lossy compression you do not get the original data back when it has been decompressed. This sort of compression would therefore NOT be suitable for anything that needed to be reproduced exactly such as software, databases or text. Example lossy formats: .mp3, .jpg, .gif, .mp4

Greater file compression = smaller file size

Lossy Compression = Reduced quality

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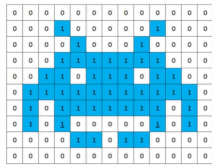
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Compression Techniques

Run-Length Encoding (lossless)

Run-length encoding utilises duplicate data and stores it in a more efficient way. To store the

image below without compression would require information about the colour of each pixel to be stored individually. An alternative solution is to group contiguous pixels of the same colour together, thereby reducing the need to store repetitive data.



The first 16 pixels are all the same colour.

Instead of using 16 bits to store each pixel individually, we could use 6 bits... the first 5 bits could represent the number of pixels (10000) and the last pixel the colour (0).

We could then repeat this process for the entire image... Obviously this sort of compression will work better on larger images that use more than 2 colours!

Dictionary Encoding (lossless)

Dictionary encoding algorithms create an index of data items, which requires less space to store than the original data.

e.g.

"ASK NOT WHAT YOUR COUNTRY CAN DO FOR YOU;
ASK WHAT YOU CAN DO FOR YOUR COUNTRY"

Index	1	2	3	4	5	6	7	8	9
Word	ask	not	what	can	do	for	you	your	country

Could be stored as: 12389456713745689 (well, its binary equivalent!)

When decompressed, the numbers can be replaced by the original values. Obviously, the index will need to be stored alongside the data, but on larger files, this will still use significantly less storage than storing the entire file without any compression.

Why Compress?

Reducing file size means:

- Smaller size on disk, therefore more disk space to store other data.
- Download speeds are increased, which can save on data allowances, improve performance of web pages and reduce internet traffic.
- Real-time streaming of video & audio without buffering

